Front Office Ledger

A decision-level accounting system for MLB front offices. Built end to end by Sean Zerillo.

Live site: **szerillo.github.io/front-office-ledger/prototype** · Code and data: **github.com/szerillo/front-office-ledger** · Contact: **sean.zerillo@gmail.com**

Every front-office argument eventually reduces to the same missing document: the ledger. Who actually made the decisions, what did each one return against what it should have returned, and how much of the result was process versus luck? I built that document for all thirty active MLB regimes, as a working public product: every trade, draft pick, free-agent signing, extension, waiver claim, and international signing since each executive took office, graded against the market expectation for that transaction type, attributed to the regime that made it, and clickable down to the individual decision. Nothing is a vibe: every number is computed from public data by a versioned, documented methodology, and the parts that are approximations say so on the page.

What the system does

Six graded acquisition channels. Trades are scored from both seats over control windows (value counts while the player stays with the club), with each delivered season weighted by win-curve leverage at the receiving club and discounted 10% per year to the deal date. Draft picks grade against a maturity-adjusted slot-expectation curve. Free agency grades realized value net of contract cost against 1,630 real contracts, with qualifying-offer signings charged their forfeited draft compensation and deferral-heavy deals costed at CBT present value. Extensions grade against the extension market for the player's service class, because the pre-arb discount is structural and the skill is choosing whom to lock up. International signings grade against their bonus-pool class. Waiver claims grade against zero, the cleanest read on pro scouting.

Two eras, one engine. Beyond the 30 active regimes, the same scorer runs over 110 archive regimes (1998-2024) with their actual start and end dates: the complete Beane and Sabean windows, Epstein in both cities, Gillick in three, Cashman's full 29-year seat. Careers are comparable across eras on identical accounting, and every archive card carries the same clickable decision ledgers, market comps, and transaction trees as the active ones.

Three top-level grades per regime. FOR (decision quality vs expectation), DEV (player development: prospects marked to market yearly on a rank-to-surplus-wins curve built from sixteen seasons of MLB Pipeline top-100 lists, with credit staying home when the prospect is traded and an inherited screen so new regimes do not collect predecessors' work), and Success (on-field results: W%, division titles, and full postseason detail). The gaps between the three are usually the most interesting numbers on the page.

Process separated from outcome. Every trade whose pieces appeared on a top-100 list carries a process check: the surrendered players priced at their deal-date prospect marks versus what they actually became. San Diego trading a 19-year-old Trea Turner reads as a 2.4-win process mistake and a 20.7-win outcome, which are different numbers and should be.

A forward view that never touches the grades. Open deals carry a projected final verdict from a self-built projection layer: Marcel-style regressed form walked forward on delta-method aging curves computed from the project's own 28,000-player value cache, prospects at their Pipeline-rank expectation. Backtested at r = 0.62-0.67 against actual next-season WAR, competitive with public systems, and quarantined: realized

value is the ledger, the projection is the heading.

An empirical trade market. From 237 deadline rental deals classified exactly from the transaction feed: a star rental win costs about 0.8 wins of eventually-realized value, the median deadline rental costs roughly nothing that ever materializes, and over half of rental buys deliver under 0.3 wins. Boston's return for one expected year of Mookie Betts was market rate; the ledger's criticism was never the price.

A Deal Desk that does the work first. Propose any trade, signing, or extension and the engine renders a verdict from the ledger's own machinery: projected value over each piece's control (with a playing-time durability factor), win-curve leverage at the club's standing, the market rate for deals of that shape with bust rates, the nearest real comps and how they ended, and counterparty intelligence. For contracts it proposes the terms itself: a calibrated Ledger offer blending market comps (the eight nearest real contracts by form, age, and position, with realized outcomes) and the model's fair price, **backtested winter by winter on 1,196 real signings 2010-2025** using only information available at the time (typical miss 0.7 years, $3.2M of AAV; mega-deal pricing carries a pedigree gate, a pitcher scarcity premium, and convex elite pricing with regression shrink, cutting the $200M-class miss 24% and removing its bias; the residual star shortfall the market itself produces is disclosed, not tuned away). Every verdict shows the club's estimated payroll before and after (book money plus arb/pre-arb estimates) against the CBT line, and exports as a one-page deal memo. A Scenario Room replays famous decision points from the archive with only at-the-time information, adjudicates counters by deterministic market rules, and grades process separately from outcome. A Horizon view walks every roster forward through 2031 on the aging curves, bounded by control and contract years, with each club's season-by-season salary book: who falls off when, and what any proposed move does to the window.

The Offseason Simulator: the whole winter, playable and priced. Any club's actual 2026-27 offseason, staged the way the calendar flows: qualifying offers and arbitration tenders (arbitration priced backward-looking, off the platform season, so the tender decision is real), the exclusive re-sign window, then an open market where every free agent negotiates through a deterministic camp that values offers in its own preferred structure, spreading the same money over extra years is not a raise. Trades price in surplus value net of salary against an ownership capacity ceiling, so the Witt-for-Soto class of fantasy deal gets refused with the arithmetic shown, and a piece that fills the counterparty's computed roster hole is worth more in their eyes. An assistant GM reads the same engines out loud on every screen: tender and QO calls, re-sign or walk advice, negotiation reads against the ask, trade counsel with named bridge pieces from your own system, and real contract comps tagged with how each aged.

The shopping fingerprint: what each front office actually buys. Every acquisition a sitting regime has made since 2016 is joined to the player's public Statcast profile in the season the front office evaluated, z-scored within season. The persistent deviations are the regime's revealed preferences: one desk pays for velocity, another for strike-zone discipline, a third lives with swing-and-miss for power. On public aggregates it already reads out in plain language on every counterparty card; pointed at a club's internal evaluations, the same construction answers the sharper question, what do rivals pay for, and what do they undervalue that you can sell them.

Validation

Accounting check A WAR-class metric must sum to team record. Roster-summed Ledger WAR vs actual wins above a 48-win baseline, 327 team-seasons (2015-2025).	**r = 0.864** (R^2 = 0.747) slope 0.818 MAE 5.35 wins

Predictive check Does decision quality carry information about future results beyond the current record? 574 pooled regime-years across 112 active and archive regimes: net decision value per season through year t, predicting the next two seasons' W% after controlling for W% to date.	**partial r = 0.276** (95% CI 0.164 to 0.371) out-of-sample R^2 0.184 to 0.245 (grouped 5-fold CV) the effect survives on held-out regimes

Both checks are reproducible from the repository with no licensed data. The value metric (Ledger WAR v1.5) is self-built from the MLB Stats API: wOBA-based batting runs with positional adjustments, FIP-based pitching anchored to league RA9, park factors from home/away run environments (3-yr rolling, halved for the road split), defense from the SABR Defensive Index where published and Statcast fielding runs plus catcher framing elsewhere, and Statcast baserunning. Park factors (v1.5) and the archive sweep's full 25,000-player universe put the accounting check at r = 0.86 (MAE 5.3 wins) across 327 team-seasons: fuller rosters sum closer to team records, which is itself evidence the metric is additive the way WAR must be. The predictive study now carries real rigor: 574 pooled regime-years from 112 regimes, a cluster-bootstrap 95% interval on the partial correlation of [0.16, 0.37], and grouped out-of-sample cross-validation (R-squared 0.184 to 0.245 on held-out regimes). Known gaps (defense coverage thins before 2016) are disclosed on the site rather than patched quietly. Pooled regime-years overlap, so the predictive check's effect size is the claim, not its p-value.

Findings the consensus did not already have

The best draft room of the data era is Brian Sabean's San Francisco, +109 wins over slot (Posey, Bumgarner, the dynasty's core), a nose ahead of Epstein's Boston at +104; among active regimes it is Chris Antonetti's Cleveland at +44. Brian Cashman's full 29-year window tells the opposite of the public story: trades +74 and drafts +17 above slot, while the checkbook grades 66 wins below water net of cost. (His draft figure has fallen twice as the system improved: once when it caught itself crediting Gerrit Cole's free-agent-era production to the 2008 pick that never signed, and again when the archive extension put his barren 1999-2004 classes into the book. A system like this is supposed to correct itself in public.)

The Baltimore gap, quantified: elite grades on decisions and development against a D+ in results through 2026. Whether you weight the machine or the harvest is a values question; the ledger puts both numbers on one row.

The best extension in the book is Cleveland's 2017 Jose Ramirez deal at about +24 wins against the class market. Atlanta's famous 2019 pair grades merely good, because the class baseline already expects extensions to be team-friendly; the skill is picking the right player.

Organizational DNA is measurable. Measuring every player who changed orgs (performance in the two seasons after arriving vs the two before), the reputations fall out of the data unprompted: Cleveland improves defenders by about +2.5 runs per season and arms by +0.19 wins; Milwaukee, Tampa Bay, and San Diego profile as pitching labs; Texas arms regress. None of this was curated.

How titles actually get built, measured across all 29 champions 1998-2026: free agency has been a majority channel exactly once (the 2009 Yankees), and the median title core was 48% built by its sitting regime in 1998-2009 against 74% in 2010-2026. The handed-down title has not gone away, though: seven of the seventeen since 2010 came on majority-inherited cores, including three straight (2021 Atlanta, 2022 Houston, 2023 Texas, 85% Jon Daniels' roster). An earlier build read this as the inherited title going extinct; the corrected roster accounting says otherwise, and the site says so.

The system audits its own inputs. Tracing the 2009 Yankees' roster exposed that the MLB transaction feed is empty before 2009; the fix was joining the Retrosheet archive (22,697 rows, 1998-2008, mapped to MLBAM ids via the Chadwick register), which put the Beckett-for-Hanley and Cabrera trades into the book and left the validation studies unchanged. A second audit (October 2026) caught a subtler hole: the

player-value universe was built from transaction-touched players, so career-one-club stars who never hit free agency (Jose Altuve, Jose Ramirez, Salvador Perez, Ronald Acuna Jr., 133 significant players in all) were invisible; restoring them raised the accounting validation to r = 0.86 and put Perez back on the 2015 championship core. Both finds are documented, because a system that grades others must audit itself in public. Every graded trade also now carries market comps: the nearest deals by size and phase from a 5,764-deal archive, so each verdict reads against what deals of that shape cost.

Head-to-head records exist and are lopsided. Every trade between two sitting regimes, netted from each seat: some relationships run 20+ wins one-sided across a decade of deals. Counterparty selection is a skill the public conversation does not price.

What this machinery does inside a club

The public site is the free-data proof. Pointed at internal data, the same architecture is a decision-support system, and the Deal Desk tab is that system already running on the free-data versions of its inputs: **deadline pricing** (the live rental-market curve says what a win costs this July and whether an offer clears the market that actually exists), **contract pricing** (suggested terms calibrated against 1,196 real signings, class-market extension baselines, CBT-aware deferral valuation, payroll and tax proximity on every proposal), **roster-horizon planning** (the window, season by season, and what any move does to it), **counterparty preparation** (every GM's trade record, channel tendencies, buy-low habits, and development effects before the call), **self-audit** (which channels actually produce, with process separated from outcome so scouting is not punished for variance), and **regime benchmarking** for ownership. Swap the public value metric for internal projections and the public prospect ranks for pref lists and every grade sharpens; nothing in the architecture assumes public data.

How it was built

Python ingestion and scoring over the MLB Stats API and Retrosheet (282,000 transaction rows, 34,000 draft picks, 28,000 players valued), SQLite storage, a single-file front end with no frameworks, and zero scraping of licensed sources: when FanGraphs and Baseball-Reference proved off-limits to programmatic access, the answer was to build and validate an open value metric rather than take shortcuts. Supplementary archives were solved, not bought: the Pipeline top-100 history came from reverse-engineering MLB's GraphQL endpoint, prospect-holding orgs from minor-league season rosters, the SABR Defensive Index from parsing its published tables, rentals classified exactly from free-agency declarations in the feed. The methodology is versioned (v0.1 through v0.3 plus sweep v2, Ledger WAR v1.5, and the archive sweep), every revision is documented in the repository with what changed and why, and the 1873-2022 Retrosheet transaction archive is already ingested for the historical backfill. Schema and channel definitions are designed to extend to other leagues.

Honest limitations, currently disclosed on the site

Defense coverage thins before 2016 and pre-2014 seasons are bat-plus-position only. Extension class discounts are documented assumptions pending a larger book. Recent-era rental ratios are right-censored while surrendered prospects mature. Retention is graded but excluded from the composite until compensation picks are modeled, because raw leakage scales with how many stars a club employs. Archive regimes grade trade, draft, and waiver channels; their FA book and development grades await the historical contract and prospect-list backfill, and regimes seated before the 2005 transaction-feed floor are truncated and flagged with their true start. Payroll arb and pre-arb figures are estimates (league minimum pre-arb; arb years at 25/40/60% of projected market value); the durability factor is a validated heuristic (it improves projection error and bias, at a small cost in rank correlation, and both numbers are published). The 2026 postseason was in progress at this build. A system like this earns trust by labeling its own soft spots, and that principle is applied everywhere.

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